Derived Environment Effects in Assamese Vowel Harmony

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1. Main Claim

Assamese vowel harmony with **exceptional triggers** and **exceptional targets** follows in an account without direct reference to morphology in the phonology:

1. exceptional triggers=**floating features**.
2. exceptional targets=a marked structure is avoided if it is derived but preserved if it is underlying: a **gang effect in HG**.

3. Theoretical Background

- **Harmonic Grammar**

⇒ Weighted constraints (Smolensky and Legendre, 2006; Legendre et al., 1990)

- **Autosegmental** representations:

⇒ Max(F) **and** Dep(F) preserve feature specifications in correspondence theory. (McCarthy and Prince, 1995)

(9) a. Max(±F)
Assign a violation mark for every [±F] input feature without an output correspondent.

b. Dep(±F)
Assign a violation mark for every [±F] output feature without an input correspondent.

4. Analysis: Exceptional triggers

- -/ija/ and -/uwa/ bear a **floating [-low]-feature**.
- This floating features raises the stem-final /ɑ/ if associated.
- The raised vowel can now undergo regular ATR-harmony.

(10) Exceptional raising

	MaxFl	*Float	Max(low)	
s ɑ l + i j a (W=)	5	5	2	H=
a. s ɑ l i j a	-1		-1	-7
b. s ɑ l i j a		-1		-5
☞ c. s o l i j a			-1	-2

6. Possible Extension

- Only a derived long vowel in Slovak undergoes diphtongization, an underlyingly long vowel is realized faithfully (14).

→ **HG:** a marked long vowel *and* addition of a μ-association to a vowel is too much: diphtongization applies for mid vowels.

(14) PDEE in Slovak (Lubowicz, 2002)

	/piv+μ/ 'beer' Gen.Pl	/čel+μ/ 'forehead'	/dceːr+a/ 'daughter'
1. Affix-triggered V-lengthening:	piːv	čeːl	–
2. Diphtongization for mid V:	–	čiel	–
	[piːv]	[čiel]	[dceːra]

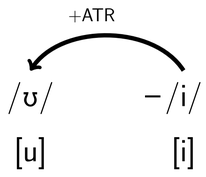
2. Data: Two Layers of Exceptionality

- All data and generalizations are from Mahanta (2008) and Mahanta (2012).
- Assamese vowel harmony spreads **[+ATR]** from **right to left** (cf. (1)) with **/ɑ/ as an opaque vowel**.

(1) Assamese regressive [+ATR]-harmony

a.	gʊl	'mix'	–i	guli	'to mix'
	pɛt	'belly'	–u	petu	'pot bellied'
b.	bɔsɔr	'year'	–i	bosori	'yearly'
	gɛrɛlɑ	'fat' (masc)	–i	gereli	'fat' (fem)

(2) *Regular ATR-Harmony*



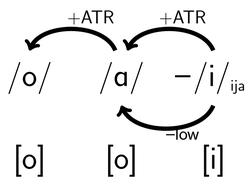
Exceptional triggers

- The exceptional suffixes -/ija/ and -/uwa/ **exceptionally raise an /ɑ/** to [o].
- /ɑ/ becomes **transparent** for [+ATR]-harmony.

(5) Exceptional triggers (Mahanta, 2012, 1121)

sɑl	'roof'	–ija	solija 'roof-ed'
dɑl	'branch'	–ija	dolija 'branch-ed'
dʰɑr	'debt'	–uwa	dʰoruwa 'debtor'
misa	'lie'	–ija	misolija 'liar'

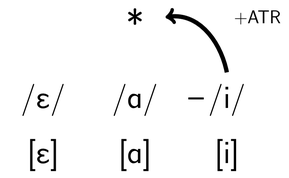
(7) *Exceptional triggers*



(3) Opaque low vowel /ɑ/ (Mahanta, 2012, 1119)

a.	kɔpah	'cotton'	–i	kɔpahi	'made of cotton'
	zʊkar	'shake'	–i	zʊkari	'shake' (Inf)
b.	lɛkʰ	'write'	–aru	lɛkʰaru	'writer'
	gɔz	'grow'	–ali	gɔzali	'sprout'

(4) *Opaque low vowel*



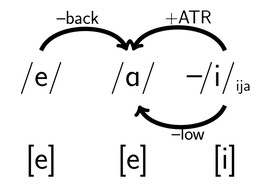
Exceptional undergoers

- Additionally, the raised /ɑ/ undergoes **exceptional frontness harmony** with a preceding mid vowel.
- The frontness harmony does not apply to underlying mid vowels, thus this is a **derived environment effect**.

(6) Exceptional undergoers (Mahanta, 2012, 1132)

a.	bɔzar	'marketplace'	–uwa	bozoruwa	'cheap'
	pɔlax	'fertiliser'	–uwa	poloxuwa	'fertile'
b.	ɛlah	'laziness'	–uwa	elehuwa	'lazy'
	kɛsa	'raw'	–uwa	keseluwa	'rawness'

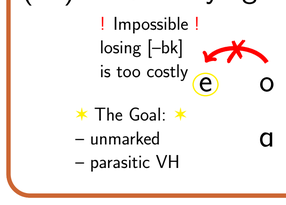
(8) *Exceptional undergoer*



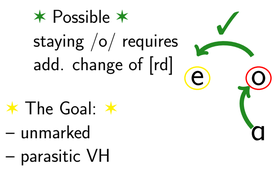
5. Analysis: Exceptional Undergoers

The Gang Effect

(11) Underlying mid V



(12) Derived mid vowel



- Exceptional harmony in derived environments is triggered by a **share constraint that requires mid vowels to agree in backness**.
- In combination with Faith(rd), *[+rd] and *[+bk] it **gangs up against** Max(bk).
- Staying back would require a violation of Faith(rd).
- Underlying mid back vowels do not front after /e/, because changing [±back] is too costly.
- There is no independent reason to violate Faith(rd).

(12) Exceptional derived frontness harmony

	Max(bk)	Share	Faith(rd)	*[+rd]	*[+bk]	H=
ɛ l ɑ h + u w ɑ (W=)	5	2	1	1	1	
a. e l o h u w a		-1	-2	-2	-3	-9
☞ b. e l e h u w a	-1			-1	-2	-8

(13) No fronting in non-derived mid vowels

	Max(bk)	Share	Faith(rd)	*[+rd]	*[+bk]	H=
x ɛ h+ɔ+t i+j a (W=)	5	2	1	1	1	
☞ a. x e h o t i j a		-1		-1	-2	-5
b. x e h e t i j a	-1		-1		-1	-7

7. Conclusion

The complex pattern of vowel harmony in Assamese involving two levels of exceptionality follows in an acocunt relying on **independently motivated mechanisms**:

- strengthening and extending the claim for **floating features** made for especially non-concatenative morphology (Zoll, 1996; Wolf, 2007).
- a **gang effect** in HG: deriving a marked structure is avoided whereas the same marked structure is preserved if underlying. This is in fact the implementation of a **Phonologically Derived Enviroment Effect**. (Kiparsky, 1973; Lubowicz, 2002; Burzio, 2011)

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